Aman Amitabh

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Education

Vellore Institute of Technology — **Vellore, Tamil Nadu, India**
Bachelor of Technology in Computer Science Engineering specializing in Internet of Things — *Aug 2023 – July 2027*
CGPA: 9.01

Mayoor Private School — **Abu Dhabi, U.A.E**
CBSE Class XII (Science Stream) — *Apr 2021 – May 2023*
10th Percentage: 90.8% | 12th Percentage: 91.6%

Summary

Computer Science student with experience building end-to-end ML and cloud-backed systems integrating IoT devices and scalable backend infrastructure. Skilled in real-time data pipelines, system design, and deploying AI-driven applications. Demonstrated leadership through technical projects and hackathons. Outside academics, an aviation enthusiast interested in complex engineered systems and an active football player who values teamwork and competitive performance.

Honors & Awards

- Achieved the Runner-Up position at SourceSprint, an open-source coding competition, recognized for technical excellence and problem solving.
- Won the Best Position Paper Award at Sonorous Youth Model United Nations conference as the Delegate of Norway in the UN General Assembly First Committee (DISEC) on the agenda Nuclear Non-Proliferation and Disarmament.
- Led a team to the Runner-Up position at aiTopia Hackathon, developing IoT and AI driven solutions aligned with the Sustainable Development Goals (SDGs).

Experience

Advanced Developers Group VIT — **Vellore, Tamil Nadu, India**
Machine Learning Lead — *Jul 2024 – Present*

- Built an Indian Monument image classifier for 19 monument classes by fine-tuning EfficientNetV2, achieving 99.69% accuracy.
- Organized and executed a 16-hour hackathon and delivered a hands-on workshop on transfer learning and API-based model inference.
- Acted as Lead Organizer and Technical Speaker for a Version Control (Git/GitHub) workshop and a GitHub bounty hunting event, engaging 120+ participants, managing event planning, task coordination and timelines while delivering the core technical sessions.

Projects

SpotFinder | *Python, YOLOv8, IoT, AWS RDS (PostgreSQL)* — *Runner-Up, aiTopia Hackathon (Feb 2025)*

- Led the development of an IoT-based smart parking system using YOLOv8 to detect parking spot occupancy edge-captured images, achieving mAP50 of 97.7% and mAP50-95 of 81%.
- Implemented an MQTT-based edge to cloud pipeline enabling occupancy updates at 1-3 FPS on resource constrained devices.
- Architected an AWS RDS (PostgreSQL) backend and developed a monitoring dashboard to store detection events and enable live occupancy tracking.

PiHome | *Python, OpenCV, IoT, MediaPipe Hands, TensorFlow Lite* — *(Apr 2025)*

- Built a gesture-controlled home automation system with a Raspberry Pi client and a Flask-based backend, offloading compute-heavy vison tasks for centralized processing.
- Designed an HTTP-based client-server pipeline to transmit frames for hand landmark extraction using MediaPipe Hands and gesture classification using a lightweight MLP with 88.28% classification accuracy.
- Mapped gesture outputs to GPIO commands on Raspberry Pi for real-time touchless appliance control.

Skills

Programming Languages: Python, Java, C, C++, HTML, CSS
Developer Tools: Git, Docker, Jupyter Notebook, Postman, VS Code
Databases: MySQL, PostgreSQL
Cloud: Microsoft Azure, AWS
AI & Machine Learning: NumPy, Pandas, Scikit-learn, TensorFlow, YOLOv8, MediaPipe, Generative AI
Core Competencies: Data Structures and Algorithms, Object Oriented Programming

Certifications

Microsoft Certified: Azure AI Engineer Associate — *(Credential ID: A82948D68646835F)*